

Question	Answer	Mark	Mark scheme	Additional guidance
7	Shown	M1	for a correct expression for the area of one face of the cube, eg x^2 or a correct expression for the surface area of the cube, eg $6 \times x^2$	No marks for $x = \sqrt{6\pi}$ without any working.
		M1	for a correct expression for the surface area of the sphere, eg $4 \times \pi \times 3^2 (= 36\pi)$	
		M1	for forming a suitable equation, eg $6 \times x^2 = 4 \times \pi \times 3^2$ or $6x^2 = "36\pi"$	$6 \times x^2 = 4 \times \pi \times 3^2$ $x^2 = 36\pi \div 6$
		A1	for completing the method to $x = \sqrt{6\pi}$ or $k = 6$	$x = \sqrt{6\pi}$